

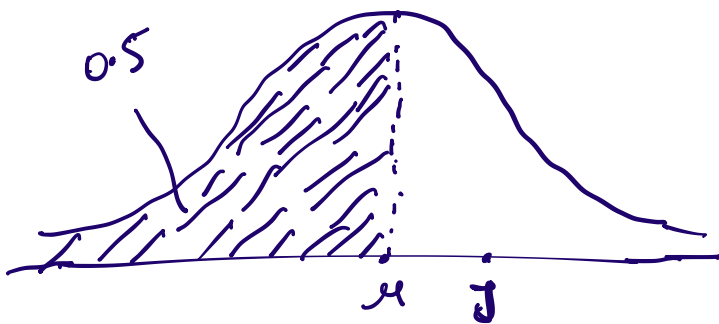
$$* \int_a^b e^{-x^2} dx = ?$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{pdf of normal dist}^n$$

Let us take  $y > 0$ :

$$CDF(y) = \int_{-\infty}^y \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx.$$

$$= \int_{-\infty}^{\mu} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx + \int_{\mu}^y \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$



$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt.$$

$$= 0.5 + \int_{\mu}^y \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \quad \text{--- (1)}$$

$$\text{put } z = \frac{(x-\mu)}{\sqrt{2}\sigma} \Rightarrow dz = \frac{dx}{\sqrt{2}\sigma}.$$

lower-limit:

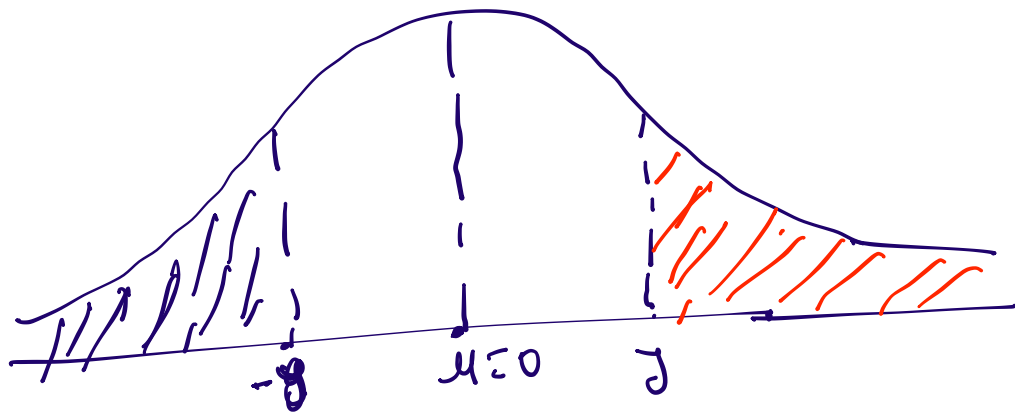
$$\text{when } x = \mu \Rightarrow z_l = \frac{\mu - \mu}{\sqrt{2}\sigma} = 0$$

upper-limit:

$$\text{when } x = y \Rightarrow z_u = \frac{y - \mu}{\sqrt{2}\sigma}.$$

$$= 0.5 + \frac{2}{\sqrt{\pi}} \int_0^{z_u} \frac{1}{\sqrt{2}} e^{-z^2} dz$$

$$CDF(y) = 0.5 + 0.5 \times \operatorname{erf}\left(\frac{y-\mu}{\sqrt{2}\sigma}\right).$$



$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$CDF(-y) = \int_{-\infty}^{-y} f(z) dz.$$

$$= 1 - \int_{-\infty}^y f(z) dz$$

$$= 1 - [0.5 + 0.5 \operatorname{erf}(y)]$$

$$\text{if } y < 0 \quad = 0.5 - 0.5 \operatorname{erf}(y)$$

required  
some correction